

Particle mixing could be geometry

An everyday explanation of the core idea behind the paper

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"The idea is that particle mixing may not be a table of numbers inserted by hand, but the consequence of a deeper geometry."

What are we trying to understand?

In particle physics there are three generations of several kinds of particles. When we look at quarks, those generations do not remain completely separate: they mix with one another according to a very specific pattern.

That pattern is normally described by a matrix full of numbers. Those numbers reproduce experiments extraordinarily well, but a deeper question remains: why do they have exactly those values and that structure?

Our work explores a different possibility: perhaps the mixing does not have to be introduced as an independent table of numbers. Perhaps it can arise as the consequence of a deeper internal geometry.

A simple picture: a visible world and a heavy sector

Imagine three "light" states representing the three family directions that we can observe. Alongside them there is a heavier sector that we do not see directly at low energies.

The three light states can participate in that heavy sector, but they do not have to do so with the same strength. One may be strongly connected, another moderately connected, and the third only weakly connected.

That inequality is essential. In the paper it is summarized by a participation profile called K . K is not yet family mixing; it simply tells us how strongly each direction participates in the heavy world.

A very simple cycle

Inside the heavy sector we introduce a minimal structure: a cyclic motion. One direction leads to the second, the second to the third, and the third returns to the first.

When that cycle is viewed through the unequal participation strengths, it is no longer perfectly uniform. Each step is weighted by how strongly its two endpoints participate.

The result is an object written in the paper as $B = K S K$. The symbol S represents the cycle and K represents the participation strengths. If all three strengths were identical, the cycle would be perfectly symmetric. If they differ, an internal asymmetry appears.

The surprise: asymmetry becomes geometry

The object B is not, by itself, a perfect rotation. From the viewpoint of the three visible states, some information appears to be lost. But that "loss" can be understood differently: the information has not been destroyed; it has moved into complementary directions that we were not looking at.

Mathematically, we find that an exact unitary completion of a generic three-state step of this kind needs three complementary directions. A $3+3$ structure therefore appears naturally: three visible directions and three complementary ones.

This does not mean that we have discovered six new physical dimensions. It means something much more precise: in this construction, six dimensions are the minimum needed to represent as perfectly conservative a step that looked non-unitary when seen only from the visible subspace.

The central result

When we construct the three-state physical plane correctly inside that $3+3$ space, a geometric connection appears. The connection tells us how the orientation of the three states changes as we move through an internal space of configurations.

The particularly striking part is that, when we use the correct oriented combination of the cycle and its reverse, that connection falls exactly inside $SU(3)$. We do not have to remove an unwanted overall phase by hand: the form of the construction itself preserves the complex volume of the three states.

$SU(3)$ is precisely the kind of mathematical structure that can produce rich rotations and mixing among three states while keeping the determinant equal to one.

What role does the inequality play?

Here we reach one of the clearest results of the work. If the three participation strengths are equal, the geometry becomes trivial: no interesting non-Abelian mixing appears.

When the three participation strengths are different, however, the cycle becomes non-normal and the geometric curvature becomes nontrivial. By studying its derivatives and commutators, we find that, generically, the geometry can generate all eight independent directions of the algebra $\mathfrak{su}(3)$.

Without formulas: the inequality between the three ways of participating in the heavy sector can be exactly what switches on the geometry needed to mix the three families.

The whole chain, without the mathematics

The idea can be summarized like this: a heavy sector has an unequal spectrum. That inequality makes the three visible directions participate differently. A minimal cycle acts on those three participation strengths. The unequal cycle produces a non-normal structure. That structure defines a geometry of planes inside a $3+3$ space. And that geometry can produce a full $SU(3)$ -type holonomy.

That is why the shortest way of describing the work is: "particle mixing could be geometry."

What have we proved - and what have we not proved?

What we have proved within the mathematical model is that heavy participation produces a well-defined contraction; that a cyclic hop generates the weighted object KSK; that a generic one-step unitary completion requires a $3+3$ space; that the anti-Hermitian construction produces a connection with exactly no extra determinant phase; and that, for sufficiently anisotropic profiles, the generated holonomy is generically $SU(3)$.

What we have not yet proved is that nature uses this exact geometry, that this mechanism by itself reproduces every observed number in the CKM matrix, or that it constitutes a complete solution of the strong CP problem.

Keeping that distinction clear is important. The paper does not say, "we have discovered the final theory of particle families." It says something more concrete and

testable: we have found a mathematical mechanism by which an unequal heavy spectrum can be transformed, through a minimal conservative construction, into an $SU(3)$ geometry capable of mixing three states.

The intuition that remains

The picture that remains is simple. Perhaps particles do not "decide" to mix because a mysterious matrix has been written into the fundamental laws. Perhaps that matrix is the visible shadow of the way three states move through a deeper geometry.

If so, what we currently call mixing parameters could be, at least in part, traces of a hidden geometric architecture.

"We are not claiming that nature has already chosen this construction. We have found a concrete way in which spectral inequality can become a geometry of mixing."

Popular-language companion to the paper "From Heavy Spectral Participation to $SU(3)$ Family Holonomy".